Result

The theoretical in-degree probability distribution for nodes of type x (where we numerically investigate the case $x \in \{a, b\}$) in the limit $\lim_{t\to\infty} p(k,t)$ for a homophilic Price model, [1, 2], given by:

$$p_{\infty,x}(k) = \begin{cases} p_{0,x}, & k = 0 \\ p_{0,x} \cdot \dfrac{B(k, \alpha_x + \gamma_x)}{B(k, \alpha_x)}, & k > 0 \end{cases} \tag{1}$$

where $B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$ where denotes the beta function.

While the underlying network model is defined only for integer-valued degrees, corresponding to $k \in \mathbb{N}$, the analytical expression admits a natural extension via analytic continuation to $k \in \mathbb{R}$. This extension is used solely for the purpose of figure generation and visual comparison with the theoretical distributions.

Parameters

The distribution parameters are:

$$\alpha_x = \frac{\mu_x}{\lambda_x} \tag{2}$$

$$\gamma_x = 1 + \frac{1}{\tilde{Z}\lambda_x} \tag{3}$$

$$p_{0,x} = \frac{1}{1 + \mu_x \tilde{Z}} \tag{4}$$

where:

- μ_x is the initial attractiveness parameter for type x nodes
- λ_x is the coupling strength for type x nodes
- $\lambda_x = f_x h_x + (1 - f_x)(1 - h_x)$
- f_x is the probability a new node added to the system is of type x.
- h_x denotes a weighting preference that a source node forms a connection with a node of the same type; $h_x \in [0, 1]$. This parameter allows us to capture, and exogenously tune, the degree of homophily in the model. For simplicity, we assign the weighting to a connection between nodes of different types to be $1 - h_x$.
- $\tilde{Z} = \frac{m}{Z}$ where $Z = g_a\lambda_a + g_b\lambda_b + f_a\mu_a + f_b\mu_b$ is the normalisation function
- $Z(t) = Z \cdot t$ is the time-dependent normalisation function
- m is the number of edges added per timestep
- g_x is the asymptotic mean in-degree for type x nodes

1 Things need to add

- Price Model Price called this the principle of cumulative advantage —the more citations you have, the more likely you are to be cited in the future. Price was inspired (Price, 1965a) by Merton's application of the Matthew effect (Merton, 1968, 1988) to the bibliometric context. This 'rich get richer' principle was new for the network context of Price but had been used in many other contexts and appears under different names [3].

- The model reappeared in 1999 as the Barab´ asi-Albert model which creates undirected networks with a power-law degree distribution. Barab´asi and Albert compared their undirected version of the model to a much wider range of networks drawn from many different contexts other than bibliometrics. It is their term, preferential attachment, which is now used to what Price called cumulative advantage.

- this whole para needs a var and citations to make sure it can be said with chest: motivated in part by the fact that this functional form is known to give rise to a scale-free degree distribution, a property widely observed in large networks [4, 2], while also incorporating both preferential attachment and node homophily; the key features of this new model.

- Homophily

- Exogenous

- Prefential attachment/ preference function

- Price's probability reasoning. essentially thats what this section is. lin khard to motivating why people should care here.

- this derivation uses long time limit tricks

2 Intro

2.1 Motivation

Abandoned intro part on motivating this research: Suppose the fraction of male AI researchers is $f \in [0, 1]$. There is an affect in the social sciences... this can be modeling using node homophily...

2.2 What is a Directed Acyclic Graph?

2.2.1 Graph Definition

To create a mathematical model of the real-world citation networks we are interested in studying, we make use of a branch of mathematics called graph theory. The term "graph" was introduced by James Joseph Sylvester in a paper published in 1878 in Nature, in which he drew an analogy between algebra and molecule diagrams [5]. A graph can be thought of as the underlying mathematical object, and a network as its real-world counterpart.

Mathematically speaking, a graph $G = (V, E)$ is an ordered pair of two sets. The first set, V, is a set of N points. Each point (in other words, each element in the set) can be labeled u_i, where $i \in \{1, 2, \ldots, N\}$, and in the language of graph theory is referred to as a node. The second set, E, is a set of pairs of these nodes, called edges [6]. Each edge is formed by choosing two nodes from V. For a given graph, if the set V contains N elements, we can, for example, write,

$$V = \{u_1, u_2, \ldots, u_{N-1}, u_N\}. \tag{5}$$

Each edge can be written as a pair (u_i, u_j) of elements of V, so the edge set E is a set of such pairs, for example,

$$E = \{(u_1, u_2), (u_k, u_1), \ldots\}, \tag{6}$$

where $k \in \{1, \ldots, N\}$.

2.2.2 A Directed Acyclic Graph

If the edges are ordered, i.e. $(u_i, u_j) \neq (u_j, u_i)$, then G is what we call a directed graph. If the edges are unordered, i.e. $(u_i, u_j) = (u_j, u_i)$, then G is an undirected graph.

In citation networks, direction arises naturally: a newer paper cites an older one. Ignoring preprints and other edge cases (such as a paper citing itself, known as a self-loop), a published paper can only cite papers that appeared earlier. This restriction imposes a temporal order on the graph, no paper can cite a paper from the future, so citation networks are naturally represented by directed graphs.

Not only does this temporal ordering mean that a citation network is a directed graph, but it also means we cannot have cycles within our graph, where node u_i forms a directed edge to u_j, which forms a directed edge to u_k, which in turn forms a directed edge back to u_i. Clearly, this would require at least one of these nodes to have an edge pointing into the future, which we have established is not possible. Since we have also removed the possibility of a self-loop, there will be no loops, in other words cycles, in this directed graph. A graph containing no cycles is called an acyclic graph. A directed graph containing no cycles is therefore called a directed acyclic graph (DAG). The citation networks we wish to model, when edge cases are removed, can therefore be described using DAGs.

To make this less abstract, a DAG is demonstrated in figure 1. This graph representation shown in figure 1, with nodes shown as circles and edges as arrows between them, is a common way to represent graphs and, in turn, the citation networks wish to model.

2.2.3 In-Degree and Citation Count

Since we are interested in modeling how many citations a paper in our network might have, it is useful to introduce some additional terminology. The in-degree of a given node, $k^{(\text{in})}$, is the total number of edges that point toward this node; in a citation network, this is the number of other papers that cite the paper represented by the node. Conversely, the out-degree of the node, $k^{(\text{out})}$, is the number of edges pointing away from the node; in a citation network, this is the number of citations that paper makes to other papers. In this analysis, since we are primarily interested in the number of citations a paper has, we denote the in-degree simply by k. In other words, $k^{(\text{in})} = k$.

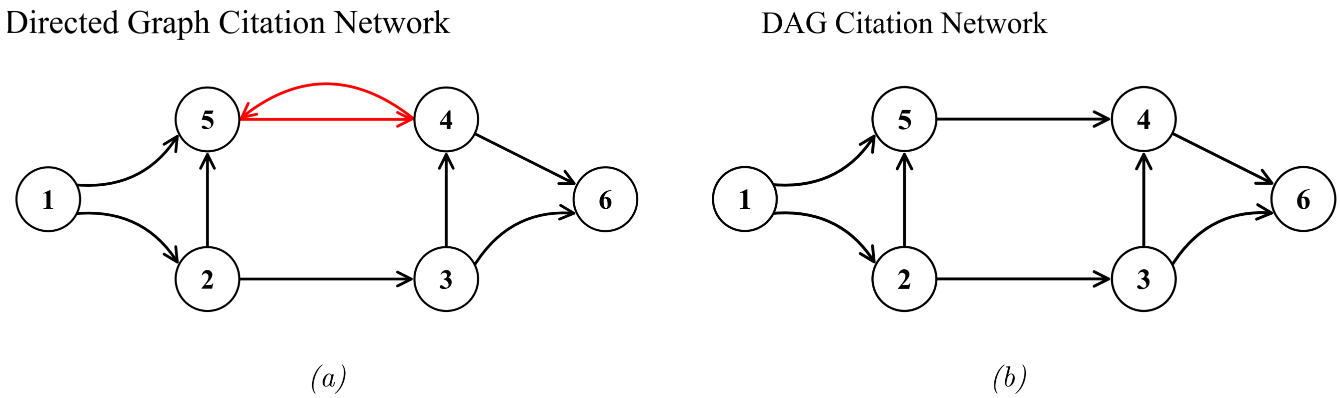


Figure 1: *(a) A citation network that is as a directed graph. Each numbered circle corresponds to a scientific publication. An arrow from paper X to paper Y indicates that X cites Y. Vertices 4 and 5, connected by reciprocal edges (shown in red), form a directed cycle. The existence of this cycle means the longest path is undefined, as one can traverse the cycle repeatedly to create arbitrarily long paths. (b) A citation network with no cycles which is a directed acyclic graph (DAG). The absence of cycles allows for well-defined longest paths. In this example, there are two longest paths: (1,2,3,4,6) and (1,2,5,4,6), each of length 4 (containing 4 edges).*

2.3 Homophily

Consider a network containing N nodes. This system can be thought of as containing two types of node: an x type node, and nodes that are not type x. It follows therefore,

$$N = N_x + N_y \tag{7}$$

where N_x is the number of type x nodes and N_y is the number of nodes that are not type x.

3 Theoretical Result Derivation

3.1 Network Model

The derivation that follows presents a new theoretical result for the probability that a node in a directed acyclic graph, designed to model real-world networks, has an in-degree of k at time t. In this graph, to model the different behaviors of different groups in real-world networks, nodes are classified into types: nodes of type x represent those that belong to a particular group, while type y nodes represent those belonging to other groups. The selection probability, $\Pi_x(k,t)$ or $\Pi_y(k,t)$, determines which node receives a new edge depending on the node's type and its current in-degree. This selection increases the selected nodes' in-degree by one.

In this derivation, $n_x(k,t)$ denotes the number of type x nodes with in-degree k at time t, where $k \in \mathbb{Z}$ and $t \in \{0\} \cup \mathbb{Z}^+$, in a directed acyclic graph constructed as follows:

At each timestep t, one new node, which can be of type x or y, is added to a preexisting directed acyclic graph containing type $\{x, y\}$ nodes. The details of how this initial network is constructed are not relevant here and are discussed in Section 4.4. The newly added node then generates

m directed edges, $m \in \{0\} \cup \mathbb{Z}^+$, to a selection of m nodes already in the graph, increasing the in-degree of these selected nodes by one. This process repeats at every timestep.

This can be seen in the two type network containing nodes of types a and b only which is presented in figure 4. Although restricting nodes to two types is arbitrary, any number of types can be reduced to two by labeling one as a focal type as x and all remaining types as y. Throughout this work, we use a two-type $\{a, b\}$ network in examples: type a nodes play the role of the focal type x, and type b nodes, being non-a, play the role of type y.

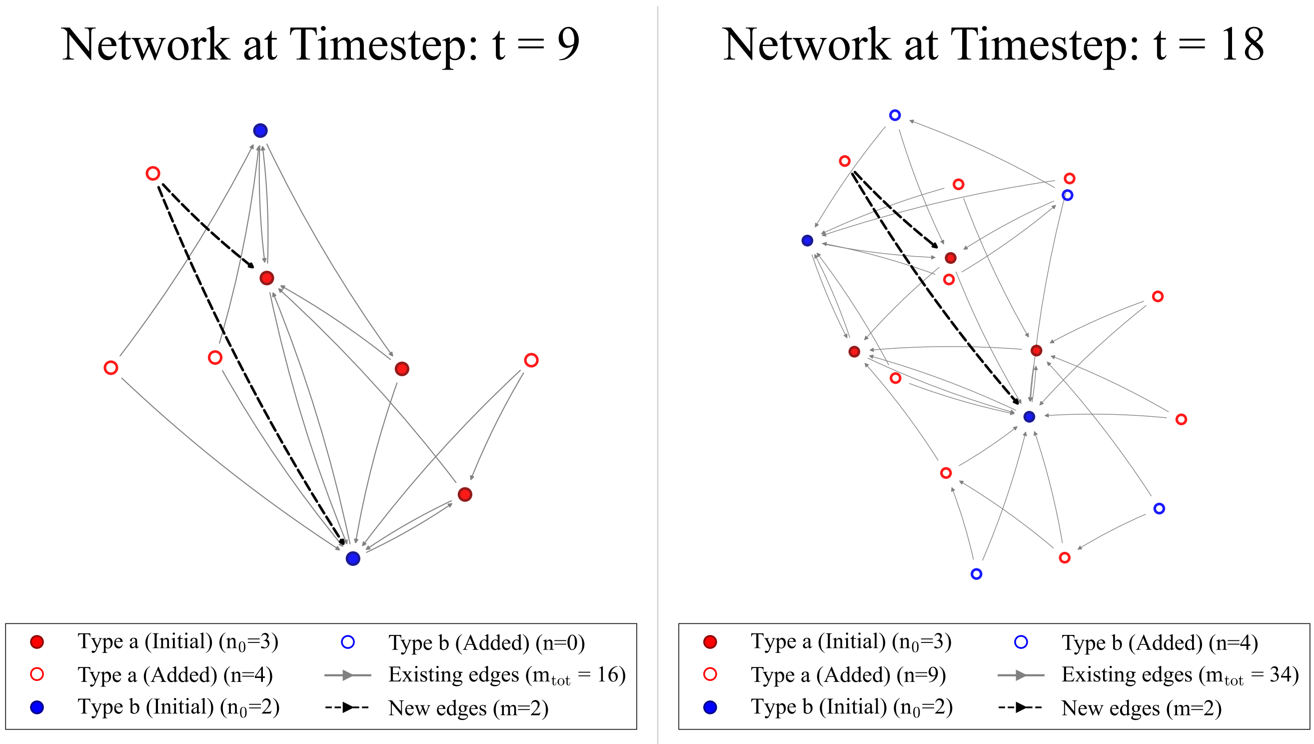


Figure 4: *Diagram of the growing directed acyclic graph (DAG) network. Both panels show the same network, which starts as an initial DAG with five nodes. The left panel shows the network at timestep* $t = 9$*, when four new nodes have been added to the original* $n_0 = 5$ *nodes. The node added at timestep* $t = 9$ *has its connections highlighted: the dashed lines indicate that it is in the process of joining the network, and the nodes these dashed arrows point to are those that have been selected probabilistically, with probability* $\Pi_x(k, t)$*, to receive a connection. The right panel shows the same process at timestep* $t = 18$*, illustrating how the network has grown further. As outlined in Section x, the placement of nodes in the diagram is arbitrary; only the pattern of connections matters. This is why the positions of the same nodes differ between the two panels.*

3.2 Difference Equation

In this model, the expected number of x type nodes with an in-degree k at time $t + 1$ can be expressed as:

$$n_x(k, t+1) = n_x(k, t) + m\Pi_x(k-1, t)n_x(k-1, t) - m\Pi_x(k, t)n_x(k, t) + \delta_{k,0} \tag{8}$$

with the convention that $n_x(k<0,t) = \Pi_x(k<0,t) = 0$. This is a difference equation, used as the starting point for deriving the probability that a node of type x (though this analysis is symmetric for type y) has in-degree k at time t [3].

The product term in equation (8),

$$m\Pi_x(k-1,t)n_x(k-1,t) \tag{9}$$

is the expected number of nodes that transition from degree $k-1$ to degree k in a single timestep, t. Since there are m new edges added by the new node created in a timestep, there are m opportunities for a given x type node to receive a new in-edge.

The $\Pi_x(k-1,t)$ term in this product term, (9), is the probability that a type x node with degree $k-1$ receives a new in-edge. We ignore cases where a newly added node creates multiple in-edges to the same existing node in a single timestep. This is because the terms in equation (8) are approximations and these multiple-edge-to-same-node events constitute an $O(m\Pi)$ correction that becomes negligible in the long-time limit (as $t \to \infty$) used in this derivation [3].

The quantity $n_x(k-1,t)$ is the number of these x type nodes with in-degree $k-1$. So, by the same reasoning, the term,

$$m\Pi_x(k,t)n_x(k,t) \tag{10}$$

in equation (8) represents the expected number of nodes leaving degree k by gaining an edge and moving to degree $k+1$.

Finally, since we are counting in-degree, the $\delta_{k,0}$ is a Kronecker delta that accounts for the case that, given the newly generated node is x type, it will be an addition to the $n(k=0,t)$ count, as the new node will necessarily have an in-degree of 0.

3.3 Preferential Attachment Probability

Now, this derivation can be thought of as an extension of the Price model [1], by accounting for multiple node types in the expression of $\Pi_x(k,t)$. This $\Pi_x(k,t)$ probability is what determines how the preferential attachment [1] and homophily [2] in our model occur. In this new model, this probability is defined as,

$$\Pi_x(k,t) = \frac{f_x h_x k + (1-f_x)(1-h_x)k + \mu_x}{Z(t)}, \tag{11}$$

Here, f_x denotes the probability that a newly added node is of type x. Since each node is either of type x or not, the probability that a node is not of type x is $1-f_x$.

3.3.1 Homophily and Initial Attractiveness

The parameter h_x denotes a weighting preference that a source node forms a connection with a node of the same type; $h_x \in [0,1]$. This parameter allows us to capture, and exogenously tune, the degree of homophily in the model.

For simplicity, we assign the weighting to a connection between nodes of different types to be $1-h_x$. This is a modeling simplification, as it assumes the attachment probability between both the same-type and the cross-type is the same. This means the homophily parameters, which we can represent by a 2×2 matrix, where the columns i correspond to the type of the newly added node and the rows j to the possible target node types, reduces to,

$$h_{ij} = \begin{bmatrix} h_{xx} & h_{xy} \\ h_{xy} & h_{yy} \end{bmatrix} = \begin{bmatrix} h_x & 1-h_x \\ 1-h_x & h_x \end{bmatrix} \tag{12}$$

The product $f_x h_x$ therefore represents the expected weighting in the preferential attachment process for same-type nodes of in-degree k to acquire a new edge, while $(1-f_x)(1-h_x)$ corresponds to the analogous weighting for different-type nodes of in-degree k. The consequences of repeating this calculation with this assumption removed are left for future work.

The constant $\mu_x \in \mathbb{R}^+$ is an exogenous parameter controlling the initial attractiveness of nodes of type x, allowing us to study how asymmetric initialisation influences network evolution. For example, by assigning one node type a substantially larger value of μ, we can investigate the extent of the resulting advantage at a given stage of the network's evolution.

3.3.2 The Normalisation Factor

It is useful to recognise that $\Pi(k,t)$ can be rewritten as,

$$\Pi_x(k,t) = \frac{\lambda_x k + \mu_x}{Z(t)} \tag{13}$$

where we have factorised the k in-degree terms and labeled the resulting factor λ_x ($\lambda_x \in \mathbb{R}^+$). The $Z(t)$ denominator is simply the normalisation factor, ensuring the probabilities of all existing nodes being targeted by the new directed edge sum to 1,

$$\sum_{i\in V_x}^{N_x} \Pi_{x,i} + \sum_{i\in V_y}^{N_y} \Pi_{y,i} = \sum_{i\in V_x}^{N_x} \frac{\lambda_x k + \mu_x}{Z(t)} + \sum_{i\in V_y}^{N_y} \frac{\lambda_y k + \mu_y}{Z(t)} = 1 \tag{14}$$

Using this constraint, we can solve for $Z(t)$.

3.4 Asymptotic Limit

If we work under the conjecture that, as t tends to infinity, both the fraction of nodes of each type and the proportion of in-edges to each type (with $E_x(t)$ denoting the number of in-edges for type x) converge to constant values, i.e.,

$$\lim_{t\to\infty} E_x(t) = \langle g_x \rangle t \quad \text{and} \quad \lim_{t\to\infty} N_x(t) = f_x t, \qquad \langle g_x \rangle, \langle g_y \rangle \in \mathbb{R}^+. \tag{15}$$

We can then rearrange (14) for $Z(t)$ in this long-time limit. To see this, let us focus on just the N_x contribution to the sum, which we can now rewrite as,

$$\sum_{i\in V_x}^{N_x} \frac{\lambda_x k + \mu_x}{Z(t)} = \frac{\lambda_x \langle g_x \rangle + \mu_x f_x}{Z(t)} \tag{16}$$

and using an identical argument for the y nodes, we can then rearrange (14) such that,

$$Z(t) = \big[\lambda_x \langle g_x \rangle + \lambda_y \langle g_y \rangle + \mu_x f_x + \mu_y f_y\big] t, \text{ with rescale } \tilde{Z} \equiv \frac{mt}{Z(t)}. \tag{17}$$

Aside from $\langle g_x \rangle$ and $\langle g_y \rangle$, this is a known analytical expression for $Z(t)$ [3]. The rescaling is purely for the sake of neater algebra. If we substitute our newly found values into (8), we get,

$$\begin{aligned} &n_x(k, t+1) - n_x(k, t) = \\ &\tilde{Z} t^{-1} \Big[\left(\lambda_x (k1) + \mu_x\right) n_x(k-1, t) - \left(\lambda_x k + \mu_x\right) n_x(k, t) \Big] + \delta_{k,0}. \end{aligned} \tag{18}$$

and since this quantity represents the expected number of nodes with in-degree k at time $t+1$, we can rewrite the equation in terms of probabilities by use of,

$$p_x(k, t) = \frac{n_x(k, t)}{N_x(t)} \tag{19}$$

which is the statement that, in the long-time limit, this fraction converges to the probability of this event. This results in,

$$\begin{aligned} &(t+1) p_x(k, t+1) - t p_x(k, t) = \\ &\tilde{Z} \Big[(\lambda_x (k-1) + \mu_x) p_x(k-1, t) - (\lambda_x (k) + \mu_x) p_x(k, t) \Big] + \delta_{k,0} \end{aligned} \tag{20}$$

Now, if we conjecture that in this long time limit there exists a stable form of each node type's probability distribution, i.e.

$$\lim_{t \to \infty} p_x(k, t) = p_{x,\infty}(k) \tag{21}$$

Then we can rewrite (20) as,

$$\frac{p_{x,\infty}(k)}{p_{x,\infty}(k, t)} = \frac{k + \frac{\mu_x}{\lambda_x} - 1}{k + \frac{\mu_x}{\lambda_x} - \frac{1}{\tilde{Z}\lambda_x}} \tag{22}$$

where it is useful to motivate, also for the sake of neat algebra,

$$\alpha_x = \frac{\mu_x}{\lambda_x} \quad \text{and} \quad \gamma_x = 1 + \frac{1}{\tilde{Z}\lambda_x}. \tag{23}$$

3.5 Gamma Function Form

If we restrict our domain to $k > 0$, there is a known result that can be used to alter the form of (22). It can be written in terms of the ratio of Gamma functions. A gamma function can be defined via the following integral [7]:

$$\Gamma(z) = \int_0^\infty t^{z-1} e^{-t} \, dt, \qquad z \in \mathbb{C}, \quad \text{where,} \quad \Re(z) > 0. \tag{24}$$

and has the following properties [3],

$$\Gamma(z+1) = z\,\Gamma(z), \qquad \Gamma(1) = 1, \qquad \text{defined for all } z \text{ except at } \begin{cases} \Gamma(1-n), & n \in \mathbb{Z}^+, \\ \Gamma(n), & n \in \mathbb{Z}^-. \end{cases} \tag{25}$$

From this definition and its properties, it is possible to rewrite equation (22) as,

$$p_{x,\infty}(k) = \frac{A_x \Gamma(k+\alpha_x)}{\Gamma(k+\alpha_x+\gamma_x)} \tag{26}$$

Therefore, if we can solve for the constant A_x, we will have our desired result. To do this, we consider equation (20) when $k = 0$. By claiming that functions with a $((k = 0) - 1)$ term are 0, this results in,

$$p_{\infty,x}(k=0) = p_{0,x} = \frac{1}{1+\tilde{Z}\mu_x} \tag{27}$$

and since equation (20) is an iterative equation valid when $k = 0$, we can use it to solve for $p_{x,\infty}(k = 1)$, which is within the domain where equation (26) is a valid. Matching the equation forms and solving for $A_x(k = 1)$, it is found that,

$$A_x(k=1) = p_{x,0}\frac{\Gamma(\alpha_x+\gamma_x)}{\Gamma(\alpha_x)}, \quad \text{and we conjecture,} \quad A_x(k) = p_{x,0}\frac{\Gamma(\alpha_x+\gamma_x)}{\Gamma(\alpha_x)}, \qquad \forall k \in \mathbb{Z}^+. \tag{28}$$

This conjecture is consistent with the known result in equation (26). A full analytical proof is left for future work. However, it can be tested numerically for selected values of k. While the stability of this result requires further analysis over a range of networks, we carried out a brief numerical test. Starting from a network with $n_0 = 5$ initial nodes, we added a further 1000 nodes. The homophily was set to $h = 0.7$, and the probability that a new node is of type a was set to $f_a = 0.6$.

In this setup, the type a and type b node counts deviated from their equation (28) predicted values, $A_x(k)$, by 3.065×10^{-8} % and -1.065×10^{-8} %, respectively, for $k \leq 166$. Within this range, fluctuations around these values were on the order of 10^{-12}. For both types, the computed values $A_a(167)$ and $A_b(167)$, and above, were infinite, likely because the relevant gamma function values could not be evaluated numerically.

The gamma function $\Gamma(z)$ grows rapidly for large arguments, and when z becomes sufficiently large (here $z \approx 167 + \alpha_{x,y} + \gamma_{x,y} \approx 169$), its value exceeds the maximum representable number in the implementation used, causing numerical overflow. Since illustrating this would amount to plotting two straight lines that abruptly terminate, we chose to omit including this numerical analysis as a figure. Acknowledging the need for future analytical work, we take this numerical analysis as provisional justification for the analytical form of $A_x(k)$, for all $k \in \mathbb{Z}^+$, seen in equation (28).

Now, when representing the A_x found in equation (26) with this equation (28) counterpart, this gives,

$$p_{x,\infty}(k) = p_0\frac{\Gamma(\alpha_x+\gamma_x)}{\Gamma(\alpha_x)}\frac{\Gamma(k+\alpha_x)}{\Gamma(k+\alpha_x+\gamma_x)}. \tag{29}$$

For the purposes of this work, this equation is a perfectly valid final result, but we simplify it further. This is done using the Beta function, which can be defined as,

$$B(a,b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}. \tag{30}$$

So, when reintroducing the $k = 0$ case, equation (29) can then be written as,

$$p_{\infty,x}(k) = \begin{cases} p_{0,x} & k = 0 \\ p_{0,x} \cdot \dfrac{B(k, \alpha_x + \gamma_x)}{B(k, \alpha_x)} & k \in \mathbb{Z}^+. \end{cases} \tag{31}$$

This, essentially, concludes the derivation.

3.6 Derivation Closing Remarks

Before moving on to analysing how well the derived equation (31) describes finite-sized realisations of these networks, it worth reiterating two points. Firstly, given the symmetrical nature of this problem,by simply swapping x for y in equation (31), we ascertain the equivalent equation for the y type nodes. Secondly, that the derivation of (31) is not a fully analytical one.

This derivation is not fully analytical because we have not yet obtained explicit expressions for $\langle g_x \rangle$ and $\langle g_y \rangle$. Aside from the $\langle g_x \rangle$ and $\langle g_y \rangle$ terms, equation (31) is an analytical solution, assuming that, as t tends to infinity, both the fraction of nodes of each type and the proportion of in-edges to each type tend to constant values.

These two unknown parameters, $\langle g_x \rangle$ and $\langle g_y \rangle$, can however be reduced to one. This is because the total number of in-degrees added at each timestep must equal the total number of out-degrees added at each timestep. This conservation of edges in the long-time limit gives,

$$\langle g_x \rangle + \langle g_y \rangle = m. \tag{32}$$

However, this still leaves one unknown parameter. While deriving analytical expressions for $\langle g_x \rangle$ and $\langle g_y \rangle$ is left for future work, numerically, we observe that the proportion of in-edges per node type do tend to constant values for large t. This can be seen in Figure 5. Moreover, as expected, these constants also sum to m.

This means that, when assessing how well equation (31) models the finite-sized simulated data, we can write $\langle g_x \rangle = m - \langle g_y \rangle$ and numerically fit the asymptotic value to which $\langle g_y \rangle$ converges. The implications of this choice are discussed later in Section 5.4.4.

This concludes the derivation. The result of this derivation is a new expression for modeling the probability that a node in a directed acyclic graph, with properties designed to capture key features (homophily and preferential attachment) seen in real-world networks, has an in-degree of k at time t.

Describe the choice made to assertain constant

4 Numerical Model

4.1 Validation Philosophy

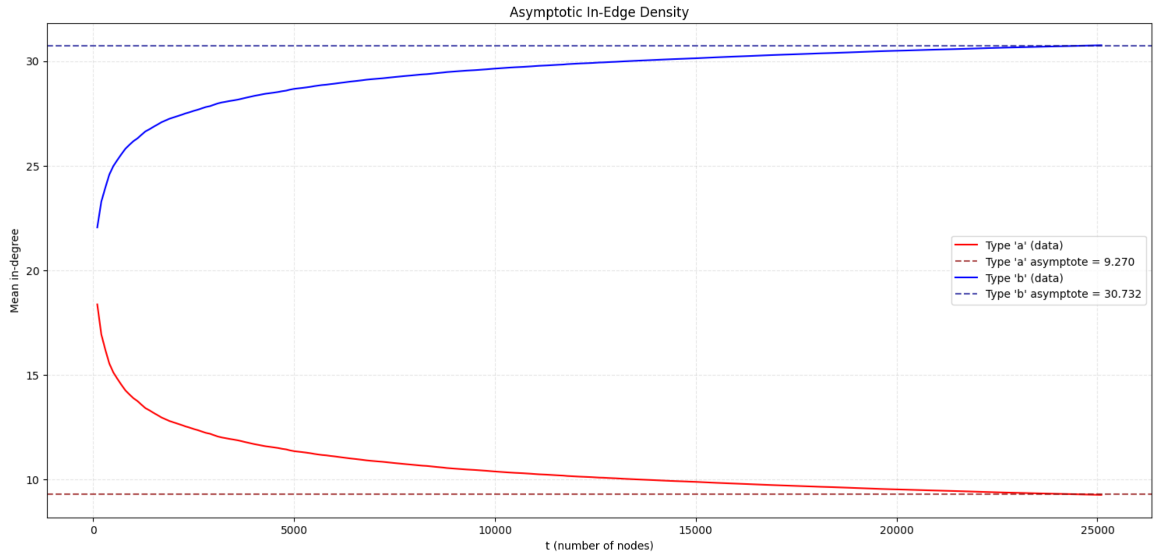


Figure 5: *Placeholder Figure. Numerical demonstration of the proportion of in-edges per node type tending to constant values as* $t \to \infty$*. This also shows these asymptotes, as expected, sum to* m*. The network model is the simplified version where* $x = a$ *and* $y = b$*, with* $f_b = 1 - f_a$ *and* $h_b = 1 - h_a$*.* $f_a = 0.4, h_a = 0.8, \mu_a = 1, \mu_b = 2, N_{initial} = 100, N_{total} = N_{initial} + 25{,}000, m = 40.$

we test this theory on finite data is real data is finite One way to assess the accuracy of our derived result, equation (31), is to compare its theoretical predictions with data generated by simulations of the process it is intended to describe. From the outset, we emphasise that the method outlined in this section provides, at best, a suggestion of agreement rather than a definitive one.

Whilst there is modern mathematical literature capable of rigorously formulating statistical tests for closely related problems to our own (e.g.[8], [9]), we find that severe difficulties arise when attempting to adapt these approaches to even slightly modified models. In particular, the assumptions required to attain this level of rigour are typically highly technical, tightly constrained, and dependent on asymptotic limit arguments that assume an infinite number of data points.

4.2 Power Law Goodness of Fit Tests

Equation (31) is not a power law. Our function, when $k > 0$, is the ratio of two Beta functions. Equation (31) may be a member of the Waring distribution family however this analysis is left for future work [10]. The key point is there is comparatively little statistical validation literature for our distribution when contrasted with the substantially richer literature on power-law fitting validation. We regard only Spierdijk and Voorneveld's work on the Yule distribution as even remotely similar [11]; however, their analysis implicitly relies on verification techniques proved to be valid for power-law distributions alongside methods such as the Pearson χ^2 test that they themselves note are inappropriate for their distribution.

What we now show is, when k is sufficiently large, (31) converges to a power law distribution. As we are considering citations that are positive real numbers, we can make use of Binet's second integral formula, which states, that if the real part of z, $z \in \mathbb{C}$, is positive then,

$$\log\Gamma(z) = \left(z - \frac{1}{2}\right)\log z - z + \frac{1}{2}\log(2\pi) + 2\int_0^\infty \frac{\arctan(t/z)}{e^{2\pi t} - 1}\,dt \tag{33}$$

For large enough k and therefore z, the integral term's contribution is negligible, so we drop this term. Now, by exponentiating (33), we have,

$$\Gamma = \sqrt{2\pi} z^{z-1/2} e^{-z} \tag{34}$$

Now, focusing on the numerator and denominator Gamma functions with k arguments in (29), and rewriting each of these using (34) before recombining the resulting expression, we obtain,

$$p_{x,\infty}(k) \propto \left[\frac{(k+\alpha_x)^{(k+\alpha_x-1/2)}}{(k+\alpha_x+\gamma_x)^{(k+\alpha_x+\gamma_x-1/2)}}\right] e^{\gamma_x} \tag{35}$$

which can be algebraically manipulated into,

$$p_{x,\infty}(k) \propto \left[\frac{(k+\alpha_x)^{-\gamma_x}}{(1+\frac{\gamma_x}{k+\alpha_x})^{-(k+\alpha_x)}(1+\frac{\gamma_x}{k+\alpha_x})^{-(\gamma_x-1/2)}}\right] e^{\gamma_x} \tag{36}$$

Now, noting,

$$\lim_{x\to\infty}\left(1 + \frac{\gamma_x}{x+\alpha_x}\right)^{x+\alpha_x} = e^{\gamma_x} \tag{37}$$

Will, for large k, be the driving term in the denominator of (36); hence,

$$p_{x,\infty}(k) \propto \frac{(k+\alpha_x)^{-\gamma_x} e^{\gamma_x}}{e^{\gamma_x}} = (k+\alpha_x)^{-\gamma_x} \tag{38}$$

Which is a power law. This of course also holds for the y counterpart version of (31). Numerically we can demonstrate this approximation and where exactly it holds well and where it holds less well.

graph and discussion of this here... As predicted, we see this holds reasonably well for larger k but not so well for smaller...

Although, as we will now discuss, these large k are in fact subject to deviations from both (38) and indeed (31) due to a phenomena we will now discuss; finite-size effects.

4.3 Finite Size Effects

No numerical model can realise an infinite number of nodes, and consequently any convergence arguments taken in the limit as t tends to infinity must be interpreted as approximations. Deviations between finite simulated networks and their infinite-size theoretical counterparts are called finite size effects [12]. Ultimately, our work is motivated by the desire to more accurately describe finite-sized, real-world citation networks, therefore, careful treatment of these finite-size effects is required. We study these finite-size effects empirically by analysing how the initial conditions of our simulation influence the resulting finite networks.

4.4 Initial Conditions

Finite-size effects in citation network models are significantly influenced by the choice of the initial conditions we set up our simulation with [13]. For some initialisations, deviations from a given theoretical citation network model persist even in the infinite-size limit [13], making numerical validation of (31) possible only by either restricting to initial conditions whose influence vanishes asymptotically or introducing explicit corrections to (31). Berset and Medo quantify this effect for a similar citation network model to ours, the classical preferential attachment model [4, 13]. We will not go into the specifics of this model beyond stating the result that, given the way this model describes citation network growth, it can be shown that the probability that a randomly selected node has degree k (when each new node introduces just a single edge) is:

$$p(k) = \frac{4}{k(k+1)(k+2)} \tag{39}$$

[14]. Berset and Medo show this result emerges for the initial nodes in the limit that t tends to infinity only when the initial network consists of a small number of sparsely interconnected nodes. They also show that when the initial nodes in this PA model have a degree above 3, this small advantage allows for the attraction of so many in-edges that these nodes never converge to the stationary degree distribution of later added nodes.

We conjecture that a similar conclusion can be reached for our network.

Obviously, simulating our network attachment process requires setting up an initial network, as in our process each new node must attach to an existing one. Since we want to investigate real citation networks, we want an initial network that balances mitigating these finite-size effects with producing structures that resemble real-world citation patterns. We now investigate this balance.

We term the initial conditions we consider in this study,

4.4.1. Minimally intrusive initialisation.

4.4.2. Variable edge density initialisation.

Initially, in this study we considered 4.4.1. This was chosen on the grounds that, for simulated networks of comparable size to real citation networks, a potentially suitable initialisation choice for reducing finite-size effects is a topology resembling one generated by the artificial citation growth process we wish to model.

To do this, we initialise the model with a fraction f_x of x-type nodes (with the remaining fraction f_y). If the model is constructed probabilistically, so that features such as node type are determined by whether model probabilities exceed a NumPy-generated random number in $[0, 1]$, then the initial network contains N_{initial} nodes with an expected,

$$\mathbb{E}[N_{x,\text{initial}}] = f_x N_{\text{initial}} \quad \text{and} \quad \mathbb{E}[N_{y,\text{initial}}] = f_y N_{\text{initial}}. \tag{40}$$

The next step is to randomly select a source node, which then connects to a target node chosen according to a specified probability distribution. For example, if we let the x type nodes have a probability of forming a connection with a node another node in the initial system,

$$\Pi_x(k,t) = \frac{\lambda_x k + \mu_x}{Y(t)}, \qquad t \in t_{initial} \tag{41}$$

then to find $Y(t)$ we follow a derivation similar to that of $Z(t)$ (17) found earlier. There are however two key differences; firstly, since the source node is already a member of the network, we need to explicitly specify that the node cannot form a connection with itself, and secondly, we cannot use the asymptotic convergence trick used in equation (15) to find a linear constant. This is because the in-degree per node-type is non-constant in $t \in t_{initial}$. To account for this first discrepancy, we introduce an additional step in the algorithm that checks the type and in-degree of the randomly selected node and removes the contribution it would otherwise make to $Y(t)$. For example, if the randomly selected node is of type x and has total in-degree k_i (where i labels the node under consideration) at time $t \in t_{\text{initial}}$, then $Y(t)$ can be written as,

$$Y(t) = \lambda_x(E_x - k_i) + (N_x - 1)\mu_x + N_y\mu_y \tag{42}$$

where a reasonable choice for the total edge counts is,

$$E^{(\text{in})}_{\text{total}} = \mathbb{E}[E_x(t)] + \mathbb{E}[E_y(t)] = \lambda_x m\, t + \lambda_y m t, \qquad t \in t_{initial} \tag{43}$$

However, this 4.4.1 initialisation has a drawback: since $t_{\text{initial}} < \infty$, the realised values will not necessarily match the expected ones. Alternatively, using a deterministic initialisation that matches the expected values exactly would restrict us to parameters that generate integer expectations. This constraint would severely restrict the model's application potential. We therefore conclude that an initialisation resembling that of our theoretical citation network either fails to align perfectly with the model, thereby allowing the initial nodes to likely retain an unfair that is non-convergent with our model, or is limited in its range of applicability.

This brings us to (4.4.2). Extrapolating from the Berset-Medo result, we conjecture that the initial network's edge density shapes finite-size effects in our model. How large are these effects, and at which edge densities do they matter? The answer tells us when our theory is informative for real networks and when finite-size effects obscure it. Because an analytical solution is difficult, we seek a statistic to compare across the number of edges per initial node and the number of initial nodes. With a well-chosen statistic, we can empirically identify initial edge densities that balance mitigating finite-size effects with producing a model suitable for analysing real-world citation networks. To proceed, we need an initial condition in which m and n_0 can vary, bringing us back to 4.4.2.

The procedure to set up 4.4.2 is identical to 4.4.1, except that the constraints enforcing the homophilic weighting mirroring are removed. Once the correct fraction is set and the rule preventing self-connections is enforced, all we do is assign the values of m and n_0 exogenously, treating these as measurable variables for comparison across different choices of these parameters. Now, with these conditions specified, we now discuss our choice of statistic used for comparison.

5 Our Test

5.1 Goodness of Fit Difficulties for Citation Networks

To understand for which initial conditions (31) provides a reasonable description of our artificial citation network, we make use of a goodness-of-fit (G.o.F.) analysis. Goodness of fit measures how well a statistical model describes a set of observations. A natural starting point is the familiar G.o.F. measure: the Pearson χ^2 statistic,

$$\chi^2(\theta_j; y_i) = \sum_{i=1}^{N} \left[\frac{y_i - f(x_i; \theta_j)}{\sigma_i} \right]^2, \tag{44}$$

where y_i are the observed data points, $f(x_i; \theta_j)$ is the theoretical prediction evaluated at x_i given parameter values θ_j, and σ_i^2 is the variance associated with each data point. In many physics applications, χ^2 offers a compact and intuitive summary of how well a theoretical distribution matches observed data.

However, the citation networks we consider are heavy-tailed. A heavy-tailed distribution of a random variable, $f(x)$, can be defined such that,

$$\int_{-\infty}^{\infty} e^{tx} df(x) = \infty \tag{45}$$

and in such cases several of the usual assumptions underlying χ^2-based inference can fail. This can be the case for the heavy-tailed power-law distribution,

$$f(x) \propto \frac{1}{x^\gamma}, \qquad x \in \mathbb{R}^+, \tag{46}$$

which is the exact form of (38). Since we observe that (38) converges to (31), we can heuristically conclude that (31) is, at least in the domain where the two equations are similar, heavy-tailed. At this point, one may naturally ask: *"If both* (31) *and* (38) *concentrate a high proportion of data at the beginning of the distribution, away from the tail, is it not overly pedantic to abandon* χ^2 *and other standard tests merely to account for an essentially negligible set of outliers?"* To answer this question, we recall from section 4.4 that we conjecture these outliers represent the region where the model is most likely to deviate. Consequently, a G.o.F. analysis for (31) that can capture tail behavior is crucial.

For (46), it has been shown that if $\gamma \leq 2$, no finite moments (no mean, no variance, etc) exist [15], while for $2 < \alpha \leq 3$ the variance diverges and the central limit theorem, in its classic formulation, does not apply, instead the mean is normally distributed with an infinite standard deviation [9]. It is important to caveat that these results apply to power-law distributions on an infinite domain. However, although our finite network simulations necessarily have bounded degrees and therefore finite empirical moments, the asymptotic regime from which both (31) and (38) are derived assumes an infinite domain; one in which the second moment may diverge. Consequently, asymptotic approximations that rely on finite second moments and independent sampling, such as the classical Pearson, χ^2 distribution, would not be theoretically justified.

This difficulty is further compounded by the dependence between the nodes in our model induced by the attachment dynamics in (11). An analysis of the extent to which the nodes in our model are dependent, that is, whether they may be regarded as weakly dependent, thereby opening avenues for alternative statistical test, is left for future work. Moreover, the standard `SciPy` χ^2 implementation assumes Poisson distribution-approximated bin counts, an assumption that does not hold for our network.

5.2 Fitting Power Laws Debate

Clearly, for our citation network, we require a G.o.F. test that is not impaired by the problematic features outlined in Section 5.1. Although the need for such a test is widely acknowledged, there remains active debate over whether an appropriate method currently exists in the literature [16, 17, 8, 18]. Our review of this literature indicates the need for a new G.o.F. procedure to adequately assess how well (31) describes our citation network. Before introducing the test we propose, it is useful to outline the statistical concepts on which it relies by discussing a closely related procedure: the influential test of Clauset et al., *Power-Law Distributions in EmpiricalData* [18].

The procedure described in [18] provides a method for generating a p-value when comparing empirical data with a fitted power-law model. However, in its standard form, this test is also inappropriate for evaluating the G.oF. of (31) on our artificial network. To both make this limitation clear and to introduce most of the statistical machinery used in our new method, we now briefly review the Clauset et al.procedure.

5.3 Clauset et al. Method

The Clauset et al. method begins by considering a set of real empirical data and a power law of form (46). The aim is to assess the G.o.F. of (46) in describing the data. Let the domain of (46) be $a \leq x_i \leq b$ where $a \in \{0 \cup \mathbb{R}^+\}$ and $b \to \infty$. The a parameter is then varied within its domain and, for each variation a Kolmogorov-Smirnov (KS) [19] statistic is computed.

5.3.1 KS Test

For the purpose of our analysis, we can think of a KS test as a statistical test that provides a quantitative indication of how likely it is that two continuous, one-dimensional probability distributions come from a given reference probability distribution. The resulting test is a computation of the following statistic,

$$D_n = sup_x|F_n(x) - F(x)| \tag{47}$$

where sup_x is the supremum of the set of distances. Intuitively, all this means is the statistic takes the largest absolute difference between the two distributions across all x values. $F_n(x)$ is the empirical cumulative distribution function (CDF),

$$F_n(x) = \frac{1}{n}\sum_{i=1}^{n} 1_{(-\infty,x]}(X_i), \quad \text{where} \quad 1_{(-\infty,x]}(X_i) = \begin{cases} 1 & \text{if } X_i \leq x \\ 0 & \text{otherwise} \end{cases} \tag{48}$$

and X_i denotes the i-th observed data value in a sample of size n. In our case, $F(x)$ can be thought of as the CDF for the theoretical distribution,

$$F(x) = \int_{-\infty}^{x} f(t)dt \tag{49}$$

where $f(x)$ is the continuous analytical probability distribution. Clearly, before the KS statistic can be computed, one must first obtain $F_n(x)$ and $F(x)$. While $F_n(x)$ can be computed directly from the data, the evaluation of $F(x)$ requires parameter estimation. To this end, as we now discuss, Clauset et al. employ maximum likelihood estimation.

5.3.2 Maximum Likelihood Estimation

Maximum likelihood estimation (MLE) treats observed independent and identically distributed (i.i.d.) random variables as draws from the same distribution with parameters θ [20]. The plausibility of a given parameter choice is quantified by a likelihood function. Under the i.i.d. assumption, this likelihood can be written as a product of the individual probabilities of each observation, $P_\theta(x_i)$:

$$L(\theta) = \prod_{i=1}^{n} P_\theta(x_i). \tag{50}$$

The larger this value, the larger the likelihood our data can be considered to have been drawn from this function. Given that the natural logarithm is monotonic, maximizing $L(\theta)$ is equivalent to maximizing the log-likelihood. So given,

$$\ell(\theta) = \log L(\theta) = \sum_{i=1}^{n} \log P_\theta(x_i), \quad \text{We compute,} \quad \nabla_\theta\, \ell(\theta) = 0. \tag{51}$$

where the solutions of the gradient equation are the MLE derived parameters. At this point, one may return to our discussions in 5.1 and ask: *"If citation network data are not necessarily independent, how can we apply MLE to fit parameters for $F(x)$ when $f(x)$ represents the network's associated probability distribution?"*

Fully addressing the tension between MLE's i.i.d. assumption and our network's inherent dependencies would require a rigorous analytical treatment of weak dependence conditions. As mentioned in section 5.1, we defer this analysis to future work. However, we can proceed on two grounds.

First, we have shown that equation (31) converges to a power law, (38), which matches the form of equation (46). From this the heuristic argument follows: Since we focus on high-degree nodes, and a sufficiently large dataset could place these nodes in the asymptotic regime, our null hypothesis, that equation (31) correctly describes high-degree nodes, implies this asymptotic description is reasonably correct. Second, Clauset et al. demonstrate empirically that MLE fitting of power laws produces reliable parameter estimates [18]. In Section 6, we validate this for our own analysis.

So, for a given value of a, a power-law model of form (46) is fitted to the empirical data with $x \geq a$ using MLE. The resulting fitted power law defines the theoretical CDF for the given $[a, \infty)$ domain. The KS statistic is then computed between this MLE parameter CDF and the empirical CDF over this domain.

The next step is to then choose the a value from the set of a values computed that generates the smallest KS statistic for this MLE fitted empirical data. We refer to this particular a as a^* and the corresponding minimum KS statistic as $D_{n,\text{emp}}(a^*)$.

5.3.3 Bootstrapped P-value

With a value for $D_{n,\mathrm{emp}}(a^*)$ in hand, Clauset et al. [18] repeat the same fitting and testing procedure on many synthetic data sets generated via a semi-parametric bootstrapping algorithm. For observations below $x_{\min}$, the smallest observed data value greater than or equal to a, they resample data from the empirical distribution; this is referred to as non-paramtetric bootstrapping. We will not use non-parametric bootstrapping in our method, hence, the details of this procedure are omitted in this paper. However, for observations above $x_{\min}$, multiple sets of synthetic data are generated from the fitted power-law model via a Monte-Carlo process; this is referred to as parametric bootstrapping. This procedure allows one to produce the set,

$$D_{n,syn} = \{D_{n,\mathrm{syn},1}(a^*)...D_{n,\mathrm{syn},N}(a^*)\}, \qquad i = \{1,..N\} \tag{52}$$

from which they compute an empirical p-value,

$$p \ = \ \frac{\#\left\{D_{n,\mathrm{syn},i}(a^*) > D_{n,\mathrm{emp}}(a^*)\right\}}{|\{D_{n,\mathrm{syn},i}(a^*)\}|}, \tag{53}$$

where $\#\{D_{n,\mathrm{syn},i}(a^*) > D_{n,\mathrm{emp}}(a^*)\}$ is the number of synthetic samples for which this inequality holds. This concludes their method.

5.3.4 Closing Remarks on Clauset et al.

Within Corral and Deluca's *Fitting and goodness-of-fit test of non-truncated and truncated power-law distributions*, they note that Clauset et al. provide no explanation of why their method should work [16]. Corral and Deluca, however, do make conjectures that begin to answer this question.

Their argument stems from the fact that, in principle, the distribution of the KS statistic is known, at least asymptotically, independently of the underlying form of the distribution, $F(x)$ [16],

$$p = 2\sum_{j=1}^{\infty}(-1)^{j-1}\exp\left[-2j^2\left(D_{emp}\sqrt{N} + 0.12D_{emp} + 0.11D_{emp}/\sqrt{N}\right)^2\right] \tag{54}$$

Since $F(x)$ is absent in (54),we can refer to (54) as distribution-free. However, this result relies on convergence of the empirical process to a mathematical object we will not discuss called a Brownian bridge [8], which in turn requires both i.i.d. observations and a continous fully specified distribution [8]. As shown in Section 5.1, our citation network violates these conditions, so equation (54) is not formally valid. Furthermore, because the MLE step re-estimates both γ_{emp} and γ_{syn} based on the data observed, the KS statistic is biased. Nonetheless, Corral and Deluca show that Eq. (54), computed via the 'probks' routine [21], behaves as an upper bound on the (53) while retaining a reasonably similar functional form [16]. From this, they argue, if the data follows a power law, increasing N (by lowering a) should reduce the KS statistic, since (54) implies $d_e \sim N^{-1/2}$ for large N. If, however, lowering a introduces a departure from power-law behavior in the data, the resulting model misclassifications may increase d_e. This argument however relies on the misclassification bias outweighing the variance reduction from the larger sample, however, there is no general reason to this.

Furthermore, it is not explicitly stated in Clauset et al. why (53) constitutes a p-value. Therefore, we put forward the following heuristic argument for why (53) can be considered as one. A p-value can be defined as,

$$p = \int_{\text{rejection region}} \rho(x)\, dx, \tag{55}$$

where $\rho(x)$ is a probability density function, defined such that the infinitesimal probability of finding the outcome in the range x to $x + dx$ is $\rho(x)\, dx$ [22]. In our setting, if this integral is approximated by a Monte Carlo sum over synthetic data sets, we can then estimate the probability mass in the rejection region as the fraction of samples where $D_{n,\text{syn},i}(a^*) > D_{n,\text{emp}}(a^*)$. This is exactly (53), hence why Clauset et al. refer to it as an empirical p-value.

Finally, within, *A practical recipe to fit discrete power-law distributions* [23], Corral demonstrates that if one uses a fully parametric procedure, data truly generated from a power-law distribution is rejected less frequently in comparison with Clauset et al.'s semi-parametric bootstrapping. After fitting the model via MLE over the selected data range, synthetic datasets are generated entirely from the fitted distribution, with appropriate normalisation over the truncated domain to ensure that the probability mass sums to one. The KS statistic is then recomputed for each synthetic dataset to obtain a more accurate result from (53).

5.4 Our New Method

Whilst the Clauset et al. method overcomes the primary drawback of the Pearson χ^2 previously considered, it still cannot be applied to our analysis. We now outline the reasons we cannot directly apply this method and, in turn, provide the new procedure we have created that can. Our method allows us to compute a heuristic G.o.F. metric for how well a heavy tailed distribution matches a dataset with non-independent data points. With this method we can assess the G.o.F. of equation (31) to our finite-sized simulation.

5.4.1 Discrete Data Correction

Our data is discrete; citation counts take integer values. This means the classical continuous KS distribution, equation (47), does not apply. As shown rigorously by Dimitrova et al. [17], when the theoretical function being investigated is discrete, the KS statistic is no longer distribution-free: its distribution depends explicitly on the underlying CDF, $F(x)$. They find, when using values derived from this continuous KS test, that the p-values produced are too large. We concede that whether this leads to inflation in the p-values produced by our method remains an open question.

What we can do however to ensure damage limitation is ensure that our theoretical function is discrete. Within [23], they use a discrete power law modeled using a Hurwitz zeta function expressed numerically via a Euler-Maclaurin expansion up a chosen cutoff as opposed to the continuous power law equation (46) [24]. For our analysis,the specifics of this cutoff is not required. Naturally, they find, using a discrete power to model discrete phenomena produces more accurate results from their G.o.F. test. Thankfully, (31) is already defined on the same set as our citation data, the positive integers, so, no correction to (31) is required.

5.4.2 Truncating from above

What we are interested in is how well the asymptotic limit derived equation (31) describes our finite-sized simulated data. To assess this, we employ the upper truncation procedure described by Corral and Deluca [16]. Specifically, for a fixed lower bound a, we progressively decrease the upper truncation threshold b, restricting the data to the interval $[a, b]$. For each such interval, the model parameters are re-estimated via MLE, and a G.o.F. test is performed using a fully parametric bootstrap.

The aim is to determine the largest interval $[a, b^*]$ on which the model cannot be rejected at a prescribed threshold p_c. Formally,

$$b^* = \max(b_i) \quad \text{s.t.} \quad p([a, b^*]) \geq p_c \tag{56}$$

where $p([a, b])$ denotes the bootstrap p-value computed under the model truncated to $[a, b]$.

The choice of p_c therefore acts as a model-selection threshold governing the maximal domain of non-rejection, compared to the conventional fixed domain hypothesis test. Corral suggests setting a safe p_c value of $p_c = 0.2$ [16]. However, as we can see from the arbitrary height below which we define the rejection region in (55), this choice, as with all significance values, is relatively arbitrary. For this reason, in section 6, we produce our analysis for a variety of p_c values and examine the sensitivity of our results to this choice.

It is important to state explicitly that as b decreases, both the empirical and bootstrap KS statistics are computed over a reduced domain. In turn, the procedure should be interpreted as identifying the largest domain on which the model remains statistically consistent with the data, rather than as establishing a global G.o.F.

Based on our analysis of the initial conditions in Section 4.4, we conjecture that the size of the domain for which $p([a, b^*]) \geq p_c$ depends on the extent to which initial-condition–driven finite-size effects distort the highest-degree nodes. In particular, if deviations from the asymptotic form are concentrated in the extreme tail, we expect upper truncation to reveal a breakdown of the model in this region.

5.4.3 KS Reweighting Section

Beset and Medo demonstrate that, when a data sample is large enough, the initial network is dense and the initial network exceeds a certain small size, the reweighted KS statistic is more suitable for statistical analysis of a truncated power-law distribution [13]. So, as opposed using equation (47), we use the reweighted,

$$D_n = \frac{sup_x|F_n(x) - F(x)|}{\sqrt{F(x)(1 - F(x))}} \tag{57}$$

Clauset et al. note that the KS statistic is known to be relatively insensitive to differences between distributions at the extreme limits of the range of x. This is because in these limits the CDFs, $F_n(x)$ and $F(x)$, necessarily tend to zero and one.

Clearly, in studying the impact of finite size effects on the high degree nodes, it is crucial our analysis is sensitive in the CDF regions where they approach one. In using the reweighted KS test we avoid this problem as the reweighted KS test is uniformly sensitive test across all $[a, b^*]$ ranges.

Thankfully, Clauset et al. note, for their method, the reweighted KS performs very similarly to the standard KS statistic [18].

5.4.4 Inference Problem Reframing

Another way in which the aim of our analysis differs from [16, 23, 18] is that we are not asking whether empirical data are well described by a given theoretical distribution (power laws in their case, equation (31) in ours). Rather, we ask whether a distribution derived strictly in an asymptotic limit provides an adequate description of a subset of data generated by a finite-size simulation.

This requires a subtle re-framing of the inferential problem. We have not derived a second analytic relation alongside (32) that would allow us to solve explicitly for $\langle g_x \rangle$ and $\langle g_y \rangle$, nor do we estimate these quantities via MLE. Instead, we take the values of $\langle g_x \rangle$ and $\langle g_y \rangle$ obtained from the first network instantiation and treat them as theory-implied inputs. As aforementioned, a fully analytic derivation of these quantities is deferred to future work.

Conditional on this assumption, namely, that these initial values may be treated analogously to MLE-derived parameters we then simulate Monte Carlo realisations and MLE perform curve fitting for p_0, γ_x, and β_x (and the corresponding y parameters when reversing subject type). Under this interpretation, the structure of the G.o.F. procedure remains formally similar to that of [18], even though the underlying inference objective differs.

5.4.5 method recipe

We now provide our recipe for the computation of a heuristic G.o.F. metric for how well a heavy tailed distribution matches a dataset with non-independent data points. With this method we can assess the G.o.F. of equation (31) for our finite-sized simulation.

Choose an arbitrary cutoff of p-value, p_c. For example, $p_c = 0.2$.

Let the domain of the data over which we assess the G.o.F. of equation (31) be $a \leq x_i \leq b$, where $a, b \in \{0\} \cup \mathbb{R}^+$. Throughout our analysis we fix $a = 0$, although, as shown in [16], it may in principle also be varied. Select a value b_i such that $\{b_i\} \subseteq [a, x_{max}]$, and let the largest element of this set be $b_{max} = x_{max}$, the largest simulated data point. This choice ensures that the first test is performed on the full dataset before any upper truncation is imposed.

In principle, for each b_i, the reweighted KS statistic (57) is computed. However, as we will see, it is often unnecessary to perform this computation for every interval $[a, b_i]$.

Replace $F(x)$ with the corresponding expression in (49) for the distribution being evaluated. In our case, the integral form of (49) associated with equation (31) is not analytically solvable. We therefore compute it numerically. After normalising over the truncation range $[a, b_i]$ containing k values ranging from k_{min} to k_max, (49) becomes,

$$F_{x,\infty}(k) = \frac{\sum_{i=k_{\min}}^{\min(k,k_{\max})} p_{x,\infty}(k_i)}{\sum_{i=k_{\min}}^{k_{\max}} p_{x,\infty}(k_i)} \tag{58}$$

For $F_n(x)$, following [23], we use the empirical survival function as opposed to empirical CDF,

$$F_n(x) = \frac{N_x}{N_{\{a,b\}}}, \tag{59}$$

where N_x denotes the number of observations in the truncated sample satisfying $x_i \geq x$, and $N_{\{a,b\}}$ is the total number of data points in the range $a \leq x_i \leq b$. Now, using equation (57), this results in the a set of KS statistics for each b_i, $D_{n,\text{theory}}([a, b_i])$.

Using the same data as before, we repeat the procedure to generate $F_n(x)$, but with one key change: instead of using the theoretical distribution from equation (31), we now use the same distribution fitted to our actual data. Specifically, we estimate the distribution parameters using MLE applied to the data in $[a, b_i]$. This gives us a new set of KS statistics for each interval, which we denote $D_{n,\text{MLE}}([a, b_i])$.

For each data instantiation, we collect both KS statistics into a set:

$$D_j = \{D_{n,\text{theory},j}([a, b_i]), D_{n,\text{MLE},j}([a, b_i])\} \tag{60}$$

where $j = 1, \ldots, N$ indexes the data instantiation. We repeat this procedure for N instantiations of the data.

We then compute the empirical p-value using equation (53), starting with the largest value of b_i (x_{max}) and working down in order of size.

Once we find a value b^*:

$$b^* = \max(b_i) \quad \text{such that} \quad p([a, b^*]) \geq p_c \quad \text{and} \quad b^* \in \{b_i\} \subseteq [a, x_{max}] \tag{61}$$

the p-value computed for the data in $[a, b^*]$ provides sufficient evidence that the data are consistent with the theoretical distribution. Since we search from largest to smallest domain, we stop at this first such domain. If no satisfactory domain is found across all b_i, then we conclude the data cannot be reasonably described by the distribution. This concludes the algorithm.

6 Results

7 Misc Junkyard of Half Baked Ideas

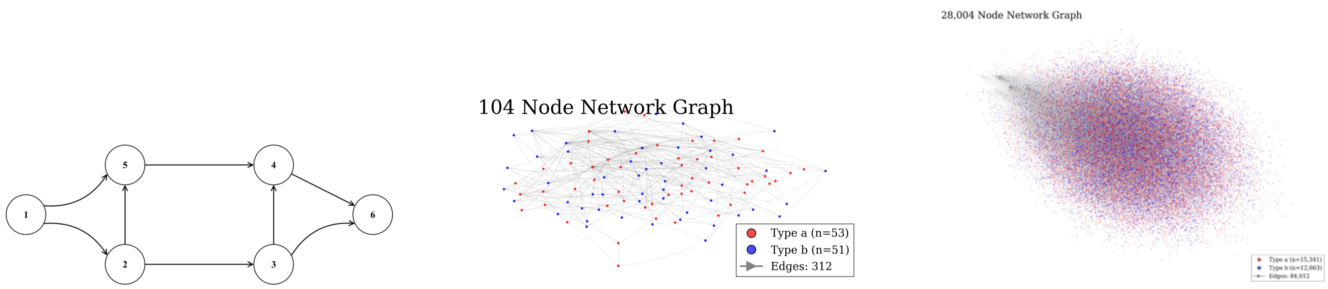
Repeat network instantiation lots of times because... Also In addition, we can obtain from the Monte Carlo simulations the uncertainty of the ML estimator by computing $\bar{\alpha}_s$, the average value of α_s, and from here its standard deviation,

$$\sigma = \sqrt{\overline{(\alpha_s - \bar{\alpha}_s)^2}},$$

where the bars indicate averaging over the N_s Monte Carlo simulations. This procedure yields good agreement with the analytical formula of Aban et al. (2006), but can be much more useful in the discrete power-law case.

$\sigma_p = \sqrt{\frac{p(1-p)}{\#(sims)}}$

One way ive been thinking about this is; this Gof makes for a compelling result if the mle-theory KS statistics were smaller as the the theory is so central for the data ranges this holds for. i.e if the theory is right, the range where the finite size effects are negligible. there are plenty of questions to ask here. i.e what does this tell us about the variance? beyond following your nose and intuition



(a) *Directed acyclic graph (DAG) diagram containing 6 nodes, labelled from one to six. Directed edges connect the nodes.*

(b) *: 104 node artificial citation network represented by homophilic directed acyclic graph. Network parameters: $n = n_0 + 100$, $m = 4$, $h_x = 0.7$, and type fraction $f_x = 0.55$.See theory section for parameter dictionary.*

(c) *28,004 node artificial citation network represented by homophilic directed acyclic graph. Network parameters: $n = n_0 + 28{,}000$, $m = 4$, $h_x = 0.7$,and type fraction $f_x = 0.55$. See theory section for parameter dictionary. Clearly visual analysis of these networks at scales of interest is not the way to go!*

Figure 6: *DAGs*

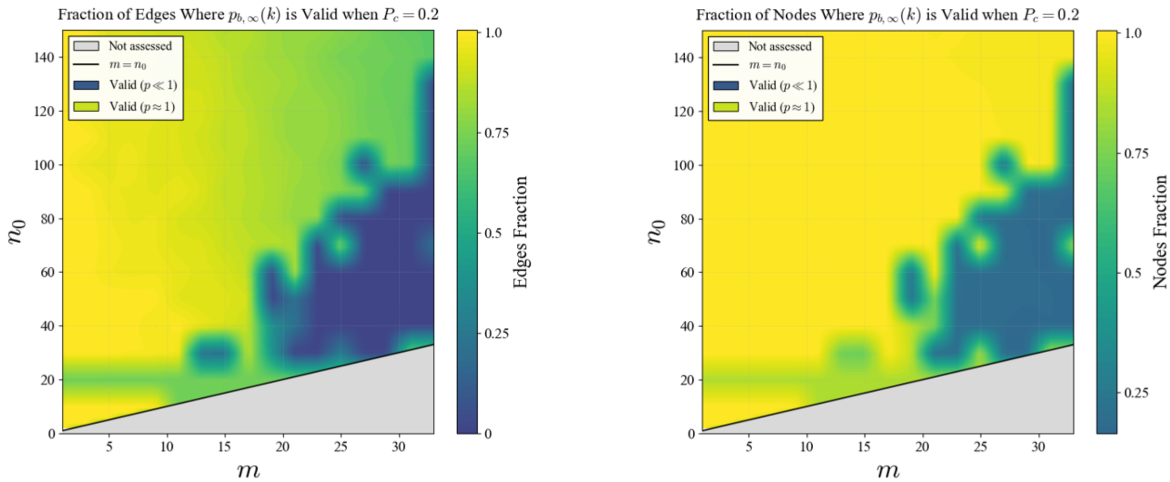
no real logic to follow one angle over another.... This motivates the appeal of carrying out this test for different amounts of tail truncation to see how much the tail is affecting our data.

Nevertheless this outlined procedure gives us a heuristic tool to measure how central our theory is for different cutoffs and this informs us how accurate our theory is. and we can perform the following experiments to get some ideas of the extent to which our theory is a good candidate to describe the data over different ranges.

By a monte Carlo approach we are referring to running N independent simulations of our network, each with identical parameters aside from the compromised seed used. Whilst the nodes in the each network are coupled, each network can be viewed as an i.i.d. data point drawn from the true distribution of the data. This therefore sets up the argument that if our theory correctly describes the data, this theory should be centrally located as, being correct would necessitate it as an unbiased estimator of the data. We must concede here it takes more than a theory being unbiased to classified as correct as this makes no comment on its fit quality as this alone gives us no indication of the data variance, however, since our i.i..d simulations are such, then the central limit theorem will apply to this set and we can compute the classic variance across the set to ascertain the per parameter uncertainty on each curve fitted function. Intuitively, we expect this variance to be well behaved when our theory accurately describes the data and less so as our data diverges from the theory. Tracking this per parameter divergence in turn will provide us with another tool to assess our conjecture regarding the finite size effects, when these are most prevalent, and what degree nodes do these affect the most.

Also, whilst we have accounted for homophily in our model, the resulting per node type probabilities of attachment that take the form of (13) are just a rescaled but the identical type of function of that used in the derivation of the Price model we base our adapted derivation off [3]. This allows us to safely use certain results in literature for more conventional preferential attachment models with this probability form.

For numerical stability, Beta functions were evaluated in log space using betaln and exponentiated after differencing, avoiding overflow and underflow arising from the super-exponential growth of



(a) *Fraction of total citations for which* (31) *passes our G.o.F test as a function of m citations each paper makes, (horizontal axis) and the n initial nodes in our model, (vertical axis). The colour scale shows the fraction of edges retained, from 0 (dark) to 1 (bright yellow), where a value of 1 indicates that all edges are within the regime where our G.o.F test accepts the theoretical distribution at the chosen threshold* $p_c = 0.2$. *10,000 nodes added*

(b) *6. Fraction of papers for which equation (1) passes our G.o.F. test as a function of m citations each paper makes, (horizontal axis) and the n initial papers in our model, (vertical axis). The colour scale shows the fraction of papers retained, from 0 (dark) to 1 (bright yellow), where a value of 1 indicates that all papers are within the regime where our G.o.F. test accepts the theoretical distribution at the chosen threshold* $p_c = 0.2$. *10,000 nodes added.*

Figure 7: *Edge fraction (left) and node fraction removed for each plot.*

the Gamma function.

Although node degrees in preferential attachment networks are dependent, the empirical degree frequencies satisfy a law of large numbers and converge to the stationary distribution. We therefore estimate parameters by maximizing the multinomial likelihood of the degree counts, which is asymptotically equivalent to maximum likelihood for the marginal degree distribution.

instead of selecting the [a,b] yourself you can automate use a possibility is to select either the interval which contains the larger number of data,(Peters et al. 2010), or the one which has the larger log-range, b/a.) but it doesn't matter how really. Its worth noting $\hat{b} = x_{max}$ if $b \geq x_{max}$ So this wont tell you anything. I.e If an acceptable fit is found, it is expected that a truncated power law, with $b \geq x_{\max}$ (where $x_{\max}$ is the largest value of x), would yield also a good fit. In fact, if b is not considered as a fixed value but as a parameter to fit, its maximum likelihood estimator when the number of data is fixed, i.e., when b is in the range $b \geq x_{\max}$, is $\hat{b} = x_{\max}$. (This is easy to see (Aban et al. 2006), just looking at Eqs. (11) and (12) for $\hat{\alpha}$, which show that $\hat{\alpha}$ increases as b approaches $x_{\max}$.) Note we cannot compare against non-truncated using AIC test as we can't fit b meaningfully. AIC is,

$$AIC = 2(\#(fitted/chosenparams) - 2Nl(\alpha_e) \tag{62}$$

But P-values are pretty arbitrary at the best of times so I think i'll graph P vs b interval considered and c.f with when certain nodes in the model are added. Accordingly, we simulate the citation growth under (11), estimate the parameters of (31) for each synthetic realisation via MLE, and compute the corresponding KS statistic. Since (31) is inherently discrete, this construction already operates in the discrete setting and mirrors the parametric bootstrap framework of [23].

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